

WASP-10b

R-2086

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_111224 · created 2026-08-10T00:32:54+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455919.6854 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1512 +/- 0.0086
Transit depth (Rp/Rs)^2	2.29 +/- 0.26 [%]
Orbital Inclination (inc)	88.43 +/- 0.71
Airmass coefficient 1 (a1)	1.001 +/- 0.011
Airmass coefficient 2 (a2)	-0.0016 +/- 0.0021
Scatter in the residuals of the lightcurve fit is	1.13 %
Best Comparison Star	#2 - [249, 225]
Optimal Method	PSF photometry
Transit Duration (day)	0.0904 +/- 0.0034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.1512 ± 0.0086 vs 0.1592 (deviation 0.93σ)
Duration fit vs published	0.0904 d vs 0.0946 d (deviation 1.23σ)
Mid-time O–C	-0.6 min
Depth SNR	17.6
Residual scatter	1.13 %

Light curve

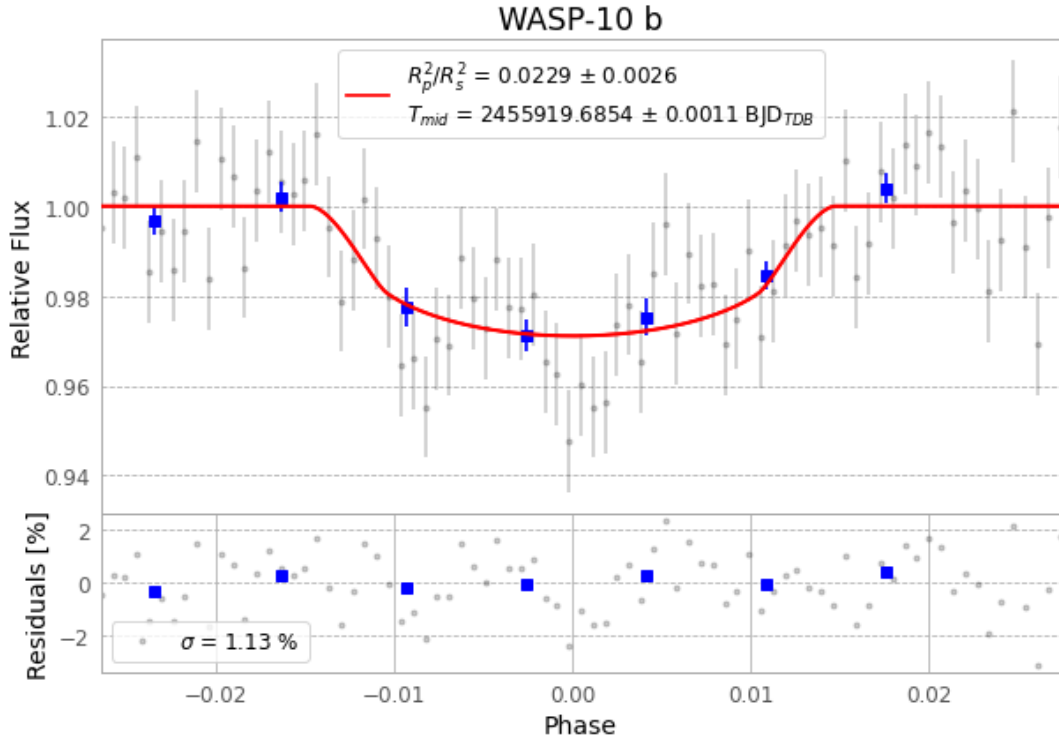


Figure 1 — FinalLightCurve_WASP-10 b_2011-12-23.png (SHA-256 928593e6aad6c486...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [254, 323], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9d1ae49f02c51042...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

81 FITS frames (53 MB), WASP-10111224022712.FITS → WASP-10111224062712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-111224.log (0d0a2f9b2c70...), FinalLightCurve_WASP-10 b_2011-12-23.png (928593e6aad6...), FinalLightCurve_WASP-10 b_2011-12-23.csv (0729c49cfc17...)

Compute resources

Wall time 172.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 00:32Z.